Chapter 2. Matrix, Calculus and Probability Tools

Matrix identities, eigenvalue inequalities, matrix calculus, optimization, convergence

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This chapter collects the tools used in later proofs (MLE, Wishart, T^2, large-sample theory). It follows the Lab Session 3 handout "Preliminaries" (used for the proofs of Results 4.11–4.13), whose item numbers are quoted in the box titles, together with the Frobenius-norm part of Lab Session 2.

1 Basic Matrix Operations

Motivation. This chapter collects the linear-algebra and probability tools used throughout, so later proofs can cite them. We start with trace, determinant, norms and continuity.

Definition 2.1: basic facts (Lab 3, Recall 1.1)

(i) For $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{p\times q}$ and $\mathbf{B} = (b_{jk}) \in \mathbb{R}^{q\times r}$, $(\mathbf{AB})_{ik} = \sum_{j=1}^{q} a_{ij}b_{jk}$.

(ii) $(\mathbf{A}^\top)_{ij} = a_{ji}$; $(\mathbf{A}^\top)^\top = \mathbf{A}$, $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$, and a 1×1 matrix (a scalar) equals its transpose.

(iii) $\mathrm{tr}(\mathbf{C}) = \sum_i c_{ii}$ for $\mathbf{C} \in \mathbb{R}^{p\times p}$, and the trace is linear: $\mathrm{tr}(a\mathbf{C} + b\mathbf{D}) = a\mathrm{tr}(\mathbf{C}) + b\mathrm{tr}(\mathbf{D})$.

(iv) $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$ for $p\times p$ matrices, so $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$ if $\mathbf{A}$ is invertible.

(v) (Leibniz) $|\mathbf{A}| = \sum_\sigma \mathrm{sgn}(\sigma)\prod_i a_{i\sigma(i)}$ over permutations σ of $\{1,\dots,p\}$, with $\mathrm{sgn}(\sigma) = (-1)^{N(\sigma)}$, $N(\sigma) = \#\{i < j : \sigma(i) > \sigma(j)\}$. Consequences: (a) $|\mathbf{A}|$ is a polynomial in the p^2 entries, so $\mathbf{A} \mapsto |\mathbf{A}|$ is continuous and infinitely differentiable, and $\{\mathbf{A} : |\mathbf{A}| \neq 0\}$ is open; (b) $|\mathbf{I}_p| = 1$, $|\mathrm{diag}(d_1,\dots,d_p)| = \prod_i d_i$, $|c\mathbf{A}| = c^p|\mathbf{A}|$.

(vi) (Adjugate) If $|\mathbf{A}| \neq 0$, $\mathbf{A}^{-1} = \mathrm{adj}(\mathbf{A})/|\mathbf{A}|$ with $\mathrm{adj}(\mathbf{A})_{ij} = (-1)^{i+j}|\mathbf{A}_{(j,i)}|$, where $\mathbf{A}_{(j,i)}$ deletes row j and column i. Each entry of $\mathbf{A}^{-1}$ is a polynomial divided by the nonzero polynomial $|\mathbf{A}|$, so $\mathbf{A} \mapsto \mathbf{A}^{-1}$ is continuous and infinitely differentiable on $\{|\mathbf{A}| \neq 0\}$ (Theorem 2.6).

(vii) (Cauchy–Schwarz for sums) $|\sum_k a_k b_k| \le (\sum_k a_k^2)^{1/2}(\sum_k b_k^2)^{1/2}$.

Definition 2.2: Frobenius norm (Lab 3, Definition 1.2; Lab 2)

For $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{p\times q}$, $\|\mathbf{A}\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$, and $\|\mathbf{x}\| = \|\mathbf{x}\|_F$ for vectors. Example: $\mathbf{M} = \begin{pmatrix} -0.1 & 0.3 \\ 0.2 & -0.2 \end{pmatrix}$ has $\|\mathbf{M}\|_F = \sqrt{0.18} \approx 0.424$. Estimation errors of the sample moments can be measured by $E_{\bar{u}} = \|\bar{\mathbf{u}} - \boldsymbol{\mu}\|_2$, $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$, $E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$, where $\boldsymbol{\mu}, \boldsymbol{\Sigma}, \mathbf{R}_0$ are the population mean, covariance and correlation matrix.

Theorem 2.3: basic identities (Lab 3, Lemma 1.3)

For $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{y} \in \mathbb{R}^q$, $\mathbf{A} \in \mathbb{R}^{p\times q}$, $\mathbf{B} \in \mathbb{R}^{q\times p}$:

(i) (bilinear form) $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_{i,j} a_{ij}x_i y_j$;

(ii) (diagonal case) if $p = q$ and $\mathbf{A} = \mathrm{diag}(a_{11},\dots,a_{pp})$, $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_i a_{ii}x_i y_i$ and $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i a_{ii}x_i^2$;

(iii) (trace) $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$;

(iv) (Frobenius norm) $\|\mathbf{A}\|_F^2 = \mathrm{tr}(\mathbf{A}^\top\mathbf{A})$;

(v) (Euclidean norm) $\|\mathbf{x}\|^2 = \mathbf{x}^\top\mathbf{x}$.

Proof. (i) $(\mathbf{Ay})_i = \sum_j a_{ij}y_j$, so $\mathbf{x}^\top\mathbf{Ay} = \sum_i x_i(\mathbf{Ay})_i = \sum_{i,j} a_{ij}x_iy_j$.

(ii) If $\mathbf{A}$ is diagonal, $a_{ij} = 0$ for $i \neq j$, so only the terms $j = i$ remain in (i).

(iii) $\operatorname{tr}(\mathbf{AB}) = \sum_i \sum_j a_{ij}b_{ji} = \sum_j \sum_i b_{ji}a_{ij} = \operatorname{tr}(\mathbf{BA})$.

(iv) $(\mathbf{A}^\top\mathbf{A})_{jj} = \sum_i (\mathbf{A}^\top)_{ji}a_{ij} = \sum_i a_{ij}^2$; sum over j.

(v) Apply (iv) to the $p \times 1$ matrix $\mathbf{x}$: $\|\mathbf{x}\|^2 = \operatorname{tr}(\mathbf{x}^\top\mathbf{x}) = \mathbf{x}^\top\mathbf{x}$. ■

Theorem 2.4: first-order expansion of the determinant (Lab 3, Lemma 1.4)

For $\mathbf{N} \in \mathbb{R}^{p\times p}$, $|\mathbf{I}_p + t\mathbf{N}| = 1 + t\operatorname{tr}(\mathbf{N}) + O(t^2)$ as $t \to 0$.

Proof. By the Leibniz formula, $|\mathbf{I}_p + t\mathbf{N}| = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i (\delta_{i\sigma(i)} + tn_{i\sigma(i)})$, where $\delta_{ii} = 1$ and $\delta_{ij} = 0$ for $i \neq j$. If σ is not the identity, $\sigma(i) \neq i$ for at least two indices, so its term is $O(t^2)$. The identity contributes $\prod_i (1 + tn_{ii}) = 1 + t\sum_i n_{ii} + O(t^2) = 1 + t\operatorname{tr}(\mathbf{N}) + O(t^2)$. ■

Definition 2.5: open sets and continuity (Lab 3, Definition 1.5)

Let V and W each be $\mathbb{R}^p$, $\mathbb{R}^{p\times q}$ or the space of $p \times p$ symmetric matrices, with distance $\|\mathbf{B} - \mathbf{A}\|_F$.

(i) $U \subset V$ is open iff for every $\mathbf{A} \in U$ there is $r > 0$ such that every $\mathbf{B} \in V$ with $\|\mathbf{B} - \mathbf{A}\|_F < r$ lies in U.

(ii) $f : U \to W$ is continuous at $\mathbf{A} \in U$ iff for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(\mathbf{B}) - f(\mathbf{A})\|_F < \epsilon$ for all $\mathbf{B} \in U$ with $\|\mathbf{B} - \mathbf{A}\|_F < \delta$; f is continuous on U if it is continuous at every point.

Theorem 2.6: entrywise continuity (Lab 3, Lemma 1.6)

With V, W as above, if every entry $F_{kl} : U \to \mathbb{R}$ of a map $F : U \to W$ is continuous at $\mathbf{A}$, then F is continuous at $\mathbf{A}$.

Proof. Let m be the number of entries of F. Given $\epsilon > 0$, choose $\delta > 0$ so that $|F_{kl}(\mathbf{B}) - F_{kl}(\mathbf{A})| < \epsilon/\sqrt{m}$ for all (k, l) whenever $\|\mathbf{B} - \mathbf{A}\|_F < \delta$. Then $\|F(\mathbf{B}) - F(\mathbf{A})\|_F^2 = \sum_{k,l}(F_{kl}(\mathbf{B}) - F_{kl}(\mathbf{A}))^2 < \epsilon^2$. ■

1.1 Block-matrix formulas

[Supplement] The following identities are not in the lab handout; they are used for conditional distributions, the Wishart theorem and likelihood ratio tests.

Theorem 2.7: block determinant and inverse [Supplement]

For $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$ with $\mathbf{A}_{22}$ invertible,

$$|\mathbf{A}| = |\mathbf{A}_{22}|\,|\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}| = |\mathbf{A}_{11}|\,|\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}| \quad (\mathbf{A}_{11} \text{ invertible}).$$

With the Schur complement $\mathbf{A}_{11\cdot 2} := \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$,

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11\cdot 2}^{-1} & -\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1} & \mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{pmatrix};$$

in particular $(\mathbf{A}^{-1})_{11} = \mathbf{A}_{11\cdot 2}^{-1}$.

Proof. Block Gaussian elimination:

$$\begin{pmatrix} \mathbf{I} & -\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \mathbf{A} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11\cdot 2} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{pmatrix}.$$

The outer factors are triangular with determinant 1, giving the determinant formula; inverting both sides gives $\mathbf{A}^{-1}$. ■

Theorem 2.8: matrix determinant lemma and Woodbury identity [Supplement]

$$|\mathbf{A} + \mathbf{u}\mathbf{v}^\top| = |\mathbf{A}|(1 + \mathbf{v}^\top \mathbf{A}^{-1}\mathbf{u}), \qquad |\mathbf{I}_p + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top \mathbf{u},$$
$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}.$$

Proof. Compute $\begin{vmatrix} \mathbf{A} & -\mathbf{u} \\ \mathbf{v}^\top & 1 \end{vmatrix}$ with both forms of the block formula: $|\mathbf{A}|(1+\mathbf{v}^\top \mathbf{A}^{-1}\mathbf{u}) = 1 \cdot |\mathbf{A}+\mathbf{u}\mathbf{v}^\top|$. For Woodbury, multiply the right side by $\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V}$ and check that the product is $\mathbf{I}$. ■

2 Eigenvalues and Positive Definiteness

Motivation. Most multivariate statistics are functions of eigenvalues of covariance matrices; their basic properties are recorded here.

Theorem 2.9: spectral decomposition and consequences (Lab 3, Theorem 2.1, Lemmas 2.2 and 2.4)

Every symmetric $p \times p$ matrix can be written $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})\mathbf{U}^\top = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal ($\mathbf{U}^\top\mathbf{U} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}_p$), real eigenvalues $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$, $\mathbf{\Lambda}(\mathbf{A}) = \operatorname{diag}(\lambda_1(\mathbf{A}), \ldots, \lambda_p(\mathbf{A}))$ and $\mathbf{A}\mathbf{u}_i = \lambda_i(\mathbf{A})\mathbf{u}_i$. Throughout, $\lambda_i(\mathbf{A})$ denotes the ith largest eigenvalue. Moreover

$$|\mathbf{A}| = \prod_i \lambda_i(\mathbf{A}), \qquad \operatorname{tr}(\mathbf{A}) = \sum_i \lambda_i(\mathbf{A}), \qquad \mathbf{A} \succ 0 \iff \text{all } \lambda_i(\mathbf{A}) > 0,$$

where $\mathbf{A} \succ 0$ (positive definite, Lab 3 Definition 2.3) means that $\mathbf{A}$ is symmetric and $\mathbf{x}^\top\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Proof. The decomposition is the spectral theorem for real symmetric matrices (any linear algebra textbook).

Determinant and trace: $|\mathbf{U}||\mathbf{U}^\top| = |\mathbf{U}\mathbf{U}^\top| = |\mathbf{I}_p| = 1$, so $|\mathbf{A}| = |\mathbf{U}||\mathbf{\Lambda}(\mathbf{A})||\mathbf{U}^\top| = |\mathbf{\Lambda}(\mathbf{A})| = \prod_i \lambda_i(\mathbf{A})$, and by Theorem 2.3(iii) $\operatorname{tr}(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top) = \operatorname{tr}(\mathbf{\Lambda}\mathbf{U}^\top\mathbf{U}) = \operatorname{tr}\mathbf{\Lambda}$.

Definiteness: with $\mathbf{z} = \mathbf{U}^\top\mathbf{x}$, $z_i = \mathbf{u}_i^\top\mathbf{x}$, Theorem 2.3(ii) gives $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \mathbf{z}^\top\mathbf{\Lambda}(\mathbf{A})\mathbf{z} = \sum_i \lambda_i(\mathbf{A})(\mathbf{u}_i^\top\mathbf{x})^2$. If all $\lambda_i(\mathbf{A}) > 0$ and $\mathbf{x} \neq \mathbf{0}$, some $\mathbf{u}_i^\top\mathbf{x} \neq 0$, so $\mathbf{x}^\top\mathbf{A}\mathbf{x} > 0$. Conversely, if $\mathbf{A} \succ 0$, $\lambda_i(\mathbf{A}) = \mathbf{u}_i^\top\mathbf{A}\mathbf{u}_i > 0$. ■

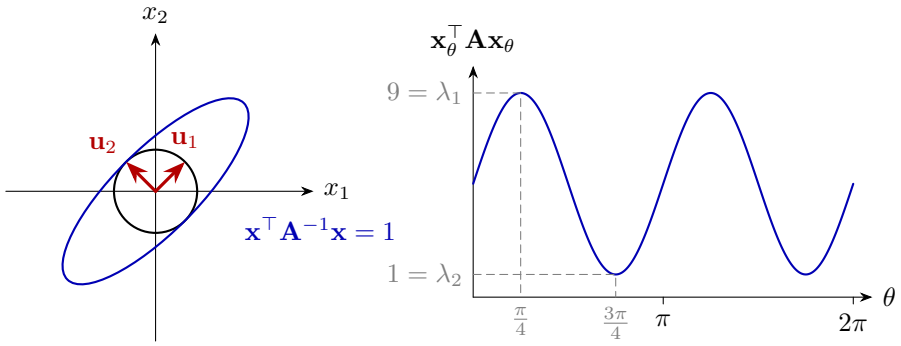
Theorem 2.10: smallest eigenvalue (Lab 3, Lemma 2.5)

For symmetric $p \times p$ matrices $\mathbf{A}, \mathbf{B}$:
(i) $\lambda_p(\mathbf{A}) = \min_{\|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{A}\mathbf{x}$;
(ii) $|\lambda_p(\mathbf{A}) - \lambda_p(\mathbf{B})| \le \|\mathbf{A} - \mathbf{B}\|_F$;
(iii) $\{\mathbf{A} \succ 0\}$ is convex and open in the space of symmetric matrices.

Proof. (i) With $\mathbf{z} = \mathbf{U}^\top \mathbf{x}$, $\|\mathbf{z}\|^2 = \mathbf{x}^\top \mathbf{U}\mathbf{U}^\top \mathbf{x} = \|\mathbf{x}\|^2 = 1$ and $\mathbf{x}^\top \mathbf{A}\mathbf{x} = \sum_i \lambda_i(\mathbf{A}) z_i^2 \ge \lambda_p(\mathbf{A})$, with equality at $\mathbf{x} = \mathbf{u}_p$.

(ii) For $\|\mathbf{x}\| = 1$, Theorem 2.3(i) and Cauchy–Schwarz give $|\mathbf{x}^\top \mathbf{A}\mathbf{x} - \mathbf{x}^\top \mathbf{B}\mathbf{x}| = |\sum_{ij}(a_{ij} - b_{ij}) x_i x_j| \le \|\mathbf{A} - \mathbf{B}\|_F (\sum_{ij} x_i^2 x_j^2)^{1/2} = \|\mathbf{A} - \mathbf{B}\|_F$. Taking the minimum over $\|\mathbf{x}\| = 1$ on both sides and using (i) gives the claim.

(iii) Convex: if $\mathbf{A}_0, \mathbf{A}_1 \succ 0$ and $s \in [0,1]$, $\mathbf{x}^\top\{(1-s)\mathbf{A}_0 + s\mathbf{A}_1\}\mathbf{x} > 0$ for $\mathbf{x} \ne \mathbf{0}$. Open: let $\mathbf{A} \succ 0$, so $\lambda_p(\mathbf{A}) > 0$. If $\mathbf{B}$ is symmetric with $\|\mathbf{B} - \mathbf{A}\|_F < \lambda_p(\mathbf{A})$, (ii) gives $\lambda_p(\mathbf{B}) \ge \lambda_p(\mathbf{A}) - \|\mathbf{A} - \mathbf{B}\|_F > 0$, so $\mathbf{B} \succ 0$; thus $r = \lambda_p(\mathbf{A})$ works in Definition 2.5(i). ■



unit circle, eigenvectors of $\mathbf{A}$

Figure. *For* $\mathbf{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ *and* $\mathbf{x}_\theta = (\cos\theta, \sin\theta)^\top$, *the quadratic form on the unit circle is* $5 + 4\sin 2\theta$. *Its maximum* $\lambda_1 = 9$ *is attained at* $\pm\mathbf{u}_1$ *(*$\theta = \pi/4$*) and its minimum* $\lambda_2 = 1$ *at* $\pm\mathbf{u}_2$ *(*$\theta = 3\pi/4$*), as in Theorem 2.10(i) and its analogue for* λ_1.

Definition 2.11: symmetric square root (Lab 3, Definition 2.6)

For $\mathbf{A} \succ 0$ with $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})\mathbf{U}^\top$, $\mathbf{A}^{1/2} := \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{1/2}\mathbf{U}^\top$ and $\mathbf{A}^{-1/2} := \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1/2}\mathbf{U}^\top$, where $\mathbf{\Lambda}(\mathbf{A})^{\pm 1/2} = \mathrm{diag}(\lambda_1(\mathbf{A})^{\pm 1/2}, \dots, \lambda_p(\mathbf{A})^{\pm 1/2})$ is well defined because all $\lambda_i(\mathbf{A}) > 0$. One spectral decomposition of $\mathbf{A}$ is fixed throughout.

Theorem 2.12: properties of the square root (Lab 3, Lemma 2.7)

For $\mathbf{A} \succ 0$:
(i) $\mathbf{A}^{1/2}$ and $\mathbf{A}^{-1/2}$ are symmetric;
(ii) $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$ and $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1}$;
(iii) $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}_p$ and $\mathbf{A}^{1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1/2}\mathbf{A}^{1/2} = \mathbf{I}_p$;
(iv) $\mathbf{A}$ is invertible, $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top$ is symmetric and $\mathbf{A}^{-1} \succ 0$, and $\lambda_i(\mathbf{A}^{-1}) = 1/\lambda_{p+1-i}(\mathbf{A})$ with eigenvector $\mathbf{u}_{p+1-i}$.

Proof. (i) Each matrix has the form $\mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{D}$ diagonal, and $(\mathbf{U}\mathbf{D}\mathbf{U}^\top)^\top = \mathbf{U}\mathbf{D}\mathbf{U}^\top$.

(ii)–(iv) Since $\mathbf{U}^\top\mathbf{U} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}_p$, $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})\mathbf{U}^\top = \mathbf{A}$, $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top$, $\{\mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top\}\mathbf{A} = \mathbf{A}\{\mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top\} = \mathbf{I}_p$, so $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})^{-1}\mathbf{U}^\top$, and $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{A}^{1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1/2}\mathbf{A}^{1/2} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}_p$. The diagonal entries of $\mathbf{\Lambda}(\mathbf{A})^{-1}$ are $1/\lambda_i(\mathbf{A}) > 0$; in

decreasing order they are $1/\lambda_p(\mathbf{A}) \geq \cdots \geq 1/\lambda_1(\mathbf{A})$, attached to $\mathbf{u}_p, \ldots, \mathbf{u}_1$. Hence $\lambda_i(\mathbf{A}^{-1}) = 1/\lambda_{p+1-i}(\mathbf{A})$, and $\mathbf{A}^{-1} \succ 0$ by Theorem 2.9. ■

3 Eigenvalue Inequalities

Motivation. PCA, canonical correlation and discriminant analysis all maximize a ratio of quadratic forms, and the normal MLE compares traces and determinants. These inequalities give a derivative-free way to optimize over covariance matrices; the trace inequality is used in the second proof of the MLE of $\boldsymbol{\Sigma}$ (Result 4.11).

Definition 2.13: Birkhoff's theorem and the rearrangement inequality (Lab 3, Recall 3.1)

(i) A $p \times p$ matrix $\mathbf{W} = (w_{ij})$ with $w_{ij} \geq 0$ and all row and column sums 1 (doubly stochastic) is a convex combination of permutation matrices.
(ii) If $a_1 \geq \cdots \geq a_p$ and $b_1 \geq \cdots \geq b_p$, then for every permutation σ, $\sum_i a_i b_{p+1-i} \leq \sum_i a_i b_{\sigma(i)} \leq \sum_i a_i b_i$.

Theorem 2.14: von Neumann's trace inequality (Lab 3, Theorem 3.2)

For symmetric $\mathbf{A} = \sum_i \lambda_i(\mathbf{A}) \mathbf{u}_i \mathbf{u}_i^\top$, $\mathbf{B} = \sum_j \lambda_j(\mathbf{B}) \mathbf{v}_j \mathbf{v}_j^\top$,

$$\sum_{i=1}^{p} \lambda_{p+1-i}(\mathbf{A}) \lambda_i(\mathbf{B}) \leq \operatorname{tr}(\mathbf{AB}) \leq \sum_{i=1}^{p} \lambda_i(\mathbf{A}) \lambda_i(\mathbf{B}).$$

The upper bound is attained when $\mathbf{v}_i = \mathbf{u}_i$ for all i and the lower bound when $\mathbf{v}_i = \mathbf{u}_{p+1-i}$ for all i (Part 1 slides, p. 176).

Proof.

Step 1 (reduction to a doubly stochastic matrix). By Theorem 2.3(iii), $\operatorname{tr}(\mathbf{AB}) = \sum_{i,j} \lambda_i(\mathbf{A}) \lambda_j(\mathbf{B}) w_{ij}$, $w_{ij} = (\mathbf{u}_i^\top \mathbf{v}_j)^2 \geq 0$. Since $\{\mathbf{u}_i\}, \{\mathbf{v}_j\}$ are orthonormal bases, $\sum_j w_{ij} = \|\mathbf{u}_i\|^2 = 1$ and $\sum_i w_{ij} = \|\mathbf{v}_j\|^2 = 1$: $\mathbf{W}$ is doubly stochastic.

Step 2 (extreme points). By Birkhoff $\mathbf{W}$ is a convex combination of permutation matrices; $\operatorname{tr}(\mathbf{AB})$ is linear in $\mathbf{W}$, so it lies between the minimum and maximum of $\sum_i \lambda_i(\mathbf{A}) \lambda_{\sigma(i)}(\mathbf{B})$ over permutations σ.

Step 3 (rearrangement). By the rearrangement inequality the maximum is at $\sigma(i) = i$ and the minimum at $\sigma(i) = p + 1 - i$.

Step 4 (equality). If $\mathbf{v}_i = \mathbf{u}_i$ (resp. $\mathbf{u}_{p+1-i}$) for all i, $\mathbf{W}$ is the identity (resp. reversal) permutation matrix and the bound is attained. ■

Theorem 2.15: an elementary inequality (Lab 3, Lemma 3.3)

$x - \log x \geq 1$ for $x > 0$, with equality iff $x = 1$.

Proof. $\phi(x) = x - \log x$ has $\phi'(x) = 1 - 1/x$, negative on $(0, 1)$ and positive on $(1, \infty)$. For $x \neq 1$ the mean value theorem gives $\phi(x) - \phi(1) = \phi'(\xi)(x - 1)$ for some ξ strictly between 1 and x, and $\phi'(\xi)(x - 1) > 0$. Hence $\phi(x) > \phi(1) = 1$. ■

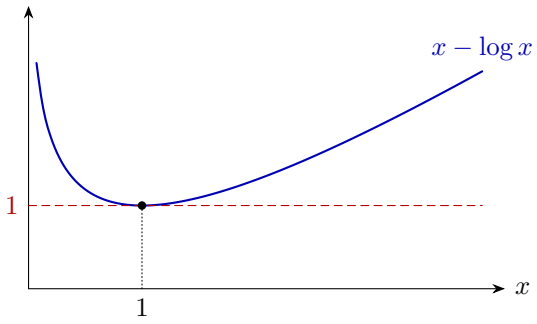


Figure. *The function $x - \log x$ decreases on $(0,1)$, increases on $(1,\infty)$ and touches its lower bound 1 only at $x = 1$. Applied to the eigenvalues of $\mathbf{\Sigma}^{-1}\mathbf{S}$ this inequality locates the normal MLE.*

4 Vector Calculus

Motivation. Maximum likelihood for the multivariate normal requires derivatives with respect to vectors and matrices, including $\log\det\mathbf{\Sigma}$. These rules are collected here.

Definition 2.16: calculus of one and several variables (Lab 3, Recall 4.1)

(i) If $g'' > 0$ on an interval, g' is strictly increasing and g is strictly convex there.
(ii) (Chain rule) If $\mathbf{y} = \mathbf{g}(\mathbf{x})$ is differentiable and f is differentiable, $\partial f(\mathbf{g}(\mathbf{x}))/\partial x_k = \sum_i (\partial f/\partial y_i)(\partial g_i/\partial x_k)$.
(iii) (Product rule) $\partial(uv)/\partial x_l = u\,\partial v/\partial x_l + v\,\partial u/\partial x_l$.

Definition 2.17: gradient, Jacobian, Hessian (Lab 3, Definition 4.2)

For differentiable $f : \mathbb{R}^p \to \mathbb{R}$, the gradient is the $p \times 1$ vector $\partial f/\partial \mathbf{x} = (\partial f/\partial x_1, \dots, \partial f/\partial x_p)^\top$, and $\partial f/\partial \mathbf{x}^\top := (\partial f/\partial \mathbf{x})^\top$. The Jacobian of a differentiable $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^q$ is the $q \times p$ matrix $\partial \mathbf{g}/\partial \mathbf{x}^\top$ with (i,j) entry $\partial g_i/\partial x_j$. For twice differentiable f the Hessian is the Jacobian of the gradient,

$$\frac{\partial^2 f}{\partial \mathbf{x}\partial \mathbf{x}^\top} := \frac{\partial}{\partial \mathbf{x}^\top}\Big(\frac{\partial f}{\partial \mathbf{x}}\Big), \qquad \Big(\frac{\partial^2 f}{\partial \mathbf{x}\partial \mathbf{x}^\top}\Big)_{ij} = \frac{\partial^2 f}{\partial x_j \partial x_i},$$

where $\partial^2 f/\partial x_j \partial x_i$ means differentiating first in x_i and then in x_j.

Theorem 2.18: symmetry of the Hessian, Schwarz's theorem (Lab 3, Theorem 4.3)

If the second partial derivatives of f are continuous, $\partial^2 f/\partial x_j \partial x_i = \partial^2 f/\partial x_i \partial x_j$ for all i, j; hence the Hessian is symmetric.

Proof. Fix $\mathbf{x}$ and $i \neq j$, and let $\Delta(h) = f(\mathbf{x} + h\mathbf{e}_i + h\mathbf{e}_j) - f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x} + h\mathbf{e}_j) + f(\mathbf{x})$.

Step 1. Applying the mean value theorem first in x_i and then in x_j, $\Delta(h) = h^2\,\partial^2 f(\boldsymbol{\xi}_h)/\partial x_j \partial x_i$ for some $\boldsymbol{\xi}_h$ with $\|\boldsymbol{\xi}_h - \mathbf{x}\| \le 2|h|$.

Step 2. Applying it in the other order, $\Delta(h) = h^2\,\partial^2 f(\boldsymbol{\eta}_h)/\partial x_i \partial x_j$ for some $\boldsymbol{\eta}_h$ with $\|\boldsymbol{\eta}_h - \mathbf{x}\| \le 2|h|$.

Step 3. Equate the two expressions, divide by h^2 and let $h \to 0$. By continuity of the second partials, $\partial^2 f(\mathbf{x})/\partial x_j \partial x_i = \partial^2 f(\mathbf{x})/\partial x_i \partial x_j$. ∎

Theorem 2.19: chain rule (Lab 3, Lemma 4.4)

Let $V \subset \mathbb{R}^p$, $U \subset \mathbb{R}^q$ be open and $\mathbf{g} : V \to U$, $f : U \to \mathbb{R}$ twice differentiable. Let $\mathbf{y} = \mathbf{g}(\mathbf{x})$ and $\mathbf{J} = \partial \mathbf{g}/\partial \mathbf{x}^\top$ $(q \times p)$. Then

(i) $\dfrac{\partial f(\mathbf{y})}{\partial \mathbf{x}} = \mathbf{J}^\top \dfrac{\partial f}{\partial \mathbf{y}}$;

(ii) $\dfrac{\partial^2 f(\mathbf{y})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{J}^\top \dfrac{\partial^2 f}{\partial \mathbf{y} \partial \mathbf{y}^\top} \mathbf{J} + \sum_{i=1}^{q} \dfrac{\partial f}{\partial y_i} \dfrac{\partial^2 g_i}{\partial \mathbf{x} \partial \mathbf{x}^\top}$;

(iii) (affine map) if $\mathbf{g}(\mathbf{x}) = \mathbf{B}\mathbf{x} + \mathbf{b}$ with fixed $q \times p$ $\mathbf{B}$ and $\mathbf{b} \in \mathbb{R}^q$, then $\mathbf{J} = \mathbf{B}$, $\dfrac{\partial f(\mathbf{y})}{\partial \mathbf{x}} = \mathbf{B}^\top \dfrac{\partial f}{\partial \mathbf{y}}$ and $\dfrac{\partial^2 f(\mathbf{y})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{B}^\top \dfrac{\partial^2 f}{\partial \mathbf{y} \partial \mathbf{y}^\top} \mathbf{B}$.

For $p = q = 1$, (i) and (ii) read $(f \circ g)' = f'(g) g'$ and $(f \circ g)'' = f''(g)(g')^2 + f'(g) g''$.

Proof. (i) Since $J_{ik} = \partial y_i / \partial x_k$, the chain rule gives $\partial f(\mathbf{y})/\partial x_k = \sum_{i=1}^{q} (\partial f/\partial y_i) J_{ik} = (\mathbf{J}^\top \partial f/\partial \mathbf{y})_k$.

(ii) Differentiate this in x_l. By the product rule and the chain rule applied to $\partial f/\partial y_i$,

$$\frac{\partial^2 f(\mathbf{y})}{\partial x_l \partial x_k} = \sum_{i,m} J_{ik} \frac{\partial^2 f}{\partial y_m \partial y_i} J_{ml} + \sum_{i=1}^{q} \frac{\partial f}{\partial y_i} \frac{\partial^2 g_i}{\partial x_l \partial x_k}.$$

The first sum is the (k,l) entry of $\mathbf{J}^\top (\partial^2 f/\partial \mathbf{y} \partial \mathbf{y}^\top) \mathbf{J}$ and the second is the (k,l) entry of $\sum_i (\partial f/\partial y_i) \partial^2 g_i / \partial \mathbf{x} \partial \mathbf{x}^\top$.

(iii) Here $g_i(\mathbf{x}) = \sum_k b_{ik} x_k + b_i$, so $J_{ik} = b_{ik}$ and every second partial of g_i is 0. ■

Theorem 2.20: gradient and Hessian of a quadratic form; Jacobians of basic maps (Lab 3, Proposition 4.5, Lemma 4.6)

For symmetric $\mathbf{A}$, $\dfrac{\partial\, \mathbf{x}^\top \mathbf{A} \mathbf{x}}{\partial \mathbf{x}} = 2\mathbf{A}\mathbf{x}$ and $\dfrac{\partial^2\, \mathbf{x}^\top \mathbf{A} \mathbf{x}}{\partial \mathbf{x} \partial \mathbf{x}^\top} = 2\mathbf{A}$. Also $\partial \mathbf{x}/\partial \mathbf{x}^\top = \mathbf{I}_p$ (the analogue of $dx/dx = 1$), $\partial(\mathbf{B}\mathbf{x}+\mathbf{b})/\partial \mathbf{x}^\top = \mathbf{B}$ for fixed $\mathbf{B} \in \mathbb{R}^{q \times p}$, $\mathbf{b} \in \mathbb{R}^q$, and, for fixed $\mathbf{x}$, $\partial(\mathbf{x} - \boldsymbol{\mu})/\partial \boldsymbol{\mu}^\top = -\mathbf{I}_p$.

Proof. By Theorem 2.3(i), $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$, so $\partial/\partial x_k = \sum_j a_{kj} x_j + \sum_i a_{ik} x_i = 2(\mathbf{A}\mathbf{x})_k$; differentiating again in x_l gives $2a_{kl}$, the (k,l) entry of $2\mathbf{A}$.

For the Jacobians: the ith component of $\mathbf{B}\mathbf{x} + \mathbf{b}$ is $\sum_k b_{ik} x_k + b_i$, whose partial in x_k is b_{ik}, the (i,k) entry of $\mathbf{B}$. Taking $\mathbf{B} = \mathbf{I}_p$, $\mathbf{b} = \mathbf{0}$ gives the identity map, and $\mathbf{B} = -\mathbf{I}_p$, $\mathbf{b} = \mathbf{x}$ (with $\boldsymbol{\mu}$ as the variable) gives the last claim. ■

5 Matrix Calculus

To optimize over covariance matrices we differentiate with respect to a matrix variable: matrix gradients give first-order conditions and directional derivatives give second-order conditions along symmetric directions. This is how the MLE of $\boldsymbol{\Sigma}$ is found in Result 4.11.

Definition 2.21: directional derivatives and matrix gradient (Lab 3, Definitions 5.1 and 5.3)

Let C be open in $\mathbb{R}^p$, $\mathbb{R}^{p\times q}$ or the space of $p\times p$ symmetric matrices, $f: C\to\mathbb{R}$, $\boldsymbol{\Theta}\in C$ and $\mathbf{H}$ a direction in the same space:

$$Df(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d}{dt} f(\boldsymbol{\Theta}+t\mathbf{H})\Big|_{t=0}, \qquad D^2 f(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d^2}{dt^2} f(\boldsymbol{\Theta}+t\mathbf{H})\Big|_{t=0}.$$

For f differentiable in the entries of $\mathbf{X}=(x_{ij})\in\mathbb{R}^{p\times q}$, the matrix gradient $\partial f/\partial\mathbf{X}$ is the $p\times q$ matrix with (i,j) entry $\partial f/\partial x_{ij}$.

Theorem 2.22: directional derivative and gradient (Lab 3, Lemma 5.2, Proposition 5.4)

(i) If $C\subset\mathbb{R}^p$ and f is twice differentiable, $Df(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \dfrac{\partial f}{\partial \boldsymbol{\mu}}$ and $D^2 f(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \dfrac{\partial^2 f}{\partial\boldsymbol{\mu}\partial\boldsymbol{\mu}^\top}\mathbf{h}$.
(ii) If f is differentiable in the pq entries of $\mathbf{X}\in\mathbb{R}^{p\times q}$, then for every $\mathbf{H}\in\mathbb{R}^{p\times q}$, $Df(\mathbf{X})[\mathbf{H}] = \operatorname{tr}\Big\{\Big(\dfrac{\partial f}{\partial \mathbf{X}}\Big)^\top \mathbf{H}\Big\} = \sum_{i,j}\dfrac{\partial f}{\partial x_{ij}} h_{ij}$. For $q=1$ this is (i).

Proof. (i) Let $g(t)=f(\boldsymbol{\mu}+t\mathbf{h})$. Theorem 2.19(iii) applied to the affine map $t\mapsto\boldsymbol{\mu}+t\mathbf{h}$ (with $\mathbf{B}=\mathbf{h}$, a $p\times 1$ matrix) gives $g'(t)=\mathbf{h}^\top\partial f(\boldsymbol{\mu}+t\mathbf{h})/\partial\boldsymbol{\mu}$ and $g''(t)=\mathbf{h}^\top\{\partial^2 f(\boldsymbol{\mu}+t\mathbf{h})/\partial\boldsymbol{\mu}\partial\boldsymbol{\mu}^\top\}\mathbf{h}$. Set $t=0$.

(ii) Each entry of $\mathbf{X}+t\mathbf{H}$ is $x_{ij}+th_{ij}$, so by the chain rule $\frac{d}{dt}f(\mathbf{X}+t\mathbf{H}) = \sum_{i,j}\frac{\partial f}{\partial x_{ij}}(\mathbf{X}+t\mathbf{H})h_{ij}$. At $t=0$ this is $\sum_{i,j} g_{ij}h_{ij}$ with $\mathbf{G}=(g_{ij})=\partial f/\partial\mathbf{X}$, and as in the proof of Theorem 2.3(iv), $\sum_{i,j} g_{ij}h_{ij} = \operatorname{tr}(\mathbf{G}^\top\mathbf{H})$. ■

Theorem 2.23: derivatives of basic matrix functions (Lab 3, Propositions 5.5 and 5.6)

For $\boldsymbol{\Sigma}\succ 0$, symmetric $\mathbf{H}$ and $p\times p$ $\mathbf{B}$:

$$D\log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}), \qquad D^2\log|\boldsymbol{\Sigma}|[\mathbf{H}] = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H}),$$

$$D\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = \operatorname{tr}(\mathbf{H}\mathbf{B}), \qquad D^2\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = 0,$$

$$\frac{\partial\log|\boldsymbol{\Sigma}|}{\partial\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}^{-1}, \qquad \frac{\partial\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})}{\partial\boldsymbol{\Sigma}} = \mathbf{B}^\top.$$

D and D^2 are linear: $D^k(a_1f_1+a_2f_2)(\boldsymbol{\Sigma})[\mathbf{H}] = a_1D^kf_1(\boldsymbol{\Sigma})[\mathbf{H}] + a_2D^kf_2(\boldsymbol{\Sigma})[\mathbf{H}]$, $k=1,2$.

Proof. Let $\mathbf{M}=\boldsymbol{\Sigma}^{-1/2}\mathbf{H}\boldsymbol{\Sigma}^{-1/2}$, symmetric by Theorem 2.12(i); write $\mathbf{M}=\mathbf{V}\boldsymbol{\Lambda}(\mathbf{M})\mathbf{V}^\top$.

Step 1. By Theorem 2.12, $\boldsymbol{\Sigma}+t\mathbf{H}=\boldsymbol{\Sigma}^{1/2}(\mathbf{I}_p+t\mathbf{M})\boldsymbol{\Sigma}^{1/2}$, and $\mathbf{I}_p+t\mathbf{M}=\mathbf{V}\{\mathbf{I}_p+t\boldsymbol{\Lambda}(\mathbf{M})\}\mathbf{V}^\top$ has eigenvalues $1+t\lambda_k(\mathbf{M})$. Hence for small $|t|$

$$\log|\boldsymbol{\Sigma}+t\mathbf{H}| = \log|\boldsymbol{\Sigma}| + \sum_{k=1}^{p}\log\{1+t\lambda_k(\mathbf{M})\}.$$

Step 2. The first and second derivatives at $t=0$ are $\sum_k\lambda_k(\mathbf{M})$ and $-\sum_k\lambda_k(\mathbf{M})^2$. Since $\mathbf{M}^2 = \mathbf{V}\boldsymbol{\Lambda}(\mathbf{M})^2\mathbf{V}^\top$, Theorems 2.9 and 2.3(iii) give $\sum_k\lambda_k(\mathbf{M}) = \operatorname{tr}\mathbf{M} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H})$ and $\sum_k\lambda_k(\mathbf{M})^2 = \operatorname{tr}(\mathbf{M}^2) = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H})$.

Step 3. $\operatorname{tr}\{(\boldsymbol{\Sigma}+t\mathbf{H})\mathbf{B}\} = \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B}) + t\operatorname{tr}(\mathbf{H}\mathbf{B})$ is linear in t. Linearity of D^k follows from that of d^k/dt^k.

Step 4 (gradients). $X \mapsto \log|\mathbf{X}|$ is differentiable in the entries on the open set $\{|\mathbf{X}| > 0\}$. For any $p \times p$ $\mathbf{H}$ let $\mathbf{N} = \boldsymbol{\Sigma}^{-1}\mathbf{H}$; then $|\boldsymbol{\Sigma} + t\mathbf{H}| = |\boldsymbol{\Sigma}||\mathbf{I}_p + t\mathbf{N}| = |\boldsymbol{\Sigma}|\{1 + t\operatorname{tr}(\mathbf{N}) + O(t^2)\}$ by Theorem 2.4, so $D\log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}) = \operatorname{tr}\{(\boldsymbol{\Sigma}^{-1})^\top \mathbf{H}\}$, as $\boldsymbol{\Sigma}^{-1}$ is symmetric. Also $D\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = \operatorname{tr}(\mathbf{H}\mathbf{B}) = \operatorname{tr}(\mathbf{B}\mathbf{H}) = \operatorname{tr}\{(\mathbf{B}^\top)^\top \mathbf{H}\}$. Comparing with Theorem 2.22(ii) for all $\mathbf{H}$ identifies the gradients. ■

Theorem 2.24: symmetric matrix variables (Lab 3, Corollary 5.7)

Let f be differentiable on an open subset of $\mathbb{R}^{p\times p}$ containing a symmetric $\mathbf{X}$, and suppose $\mathbf{G} = \partial f/\partial \mathbf{X}$ is symmetric. Then $Df(\mathbf{X})[\mathbf{H}] = 0$ for all symmetric $\mathbf{H}$ iff $\mathbf{G} = \mathbf{0}$.

Proof. If $\mathbf{G} = \mathbf{0}$, $Df(\mathbf{X})[\mathbf{H}] = \operatorname{tr}(\mathbf{G}^\top \mathbf{H}) = 0$ by Theorem 2.22(ii). Conversely, take $\mathbf{H} = \mathbf{G}$ and use Theorem 2.3(iv): $0 = Df(\mathbf{X})[\mathbf{G}] = \operatorname{tr}(\mathbf{G}^\top \mathbf{G}) = \|\mathbf{G}\|_F^2$, so $\mathbf{G} = \mathbf{0}$. ■

6 Optimization

Motivation. A critical point of a strictly convex function is its unique minimizer, some transformations of the objective do not change the optimizer, and a function of two parameters can be minimized one parameter at a time. Result 4.11 (the normal MLE) uses all three.

Theorem 2.25: convexity and unique minimizer (Lab 3, Proposition 6.1)

Let C be an open convex set in $\mathbb{R}^p$ or in the space of $p \times p$ symmetric matrices, and $f : C \to \mathbb{R}$ such that $t \mapsto f(\boldsymbol{\Theta} + t\mathbf{H})$ is twice differentiable for all $\boldsymbol{\Theta} \in C$ and all $\mathbf{H}$. Suppose $D^2 f(\boldsymbol{\Theta})[\mathbf{H}] > 0$ for every $\boldsymbol{\Theta} \in C$ and every $\mathbf{H} \neq \mathbf{0}$. Then
(i) f is strictly convex on C;
(ii) a point $\hat{\boldsymbol{\Theta}} \in C$ with $Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = 0$ for all $\mathbf{H}$ is the unique global minimizer of f on C;
(iii) for $C = \mathbb{R}^p$ and f with continuous second partials, these conditions read $\partial^2 f/\partial\boldsymbol{\Theta}\partial\boldsymbol{\Theta}^\top \succ 0$ and $\partial f(\hat{\boldsymbol{\Theta}})/\partial\boldsymbol{\Theta} = \mathbf{0}$.

Proof. (i) Take $\boldsymbol{\Theta}_0 \neq \boldsymbol{\Theta}_1$ in C, let $\mathbf{H} = \boldsymbol{\Theta}_1 - \boldsymbol{\Theta}_0$ and $g(t) = f(\boldsymbol{\Theta}_0 + t\mathbf{H})$, $t \in [0,1]$; the segment lies in C by convexity. Then $g''(t) = D^2 f(\boldsymbol{\Theta}_0 + t\mathbf{H})[\mathbf{H}] > 0$, so g is strictly convex on every segment (Definition 2.16(i)): f is strictly convex.

(ii) Let $\boldsymbol{\Theta}_0 = \hat{\boldsymbol{\Theta}}$. Then $g'(0) = Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = 0$ and g' is strictly increasing, so $g'(t) > 0$ for $t \in (0,1]$. By the mean value theorem $g(1) - g(0) = g'(\xi)$ for some $\xi \in (0,1)$, so $f(\boldsymbol{\Theta}_1) = g(1) > g(0) = f(\hat{\boldsymbol{\Theta}})$.

(iii) By Theorem 2.22(i), $D^2 f(\boldsymbol{\Theta})[\mathbf{H}] = \mathbf{H}^\top(\partial^2 f/\partial\boldsymbol{\Theta}\partial\boldsymbol{\Theta}^\top)\mathbf{H}$. The Hessian is symmetric (Theorem 2.18), so this is positive for all $\mathbf{H} \neq \mathbf{0}$ iff the Hessian is positive definite; and $Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = \mathbf{H}^\top \partial f(\hat{\boldsymbol{\Theta}})/\partial\boldsymbol{\Theta} = 0$ for all $\mathbf{H}$ iff $\partial f(\hat{\boldsymbol{\Theta}})/\partial\boldsymbol{\Theta} = \mathbf{0}$. ■

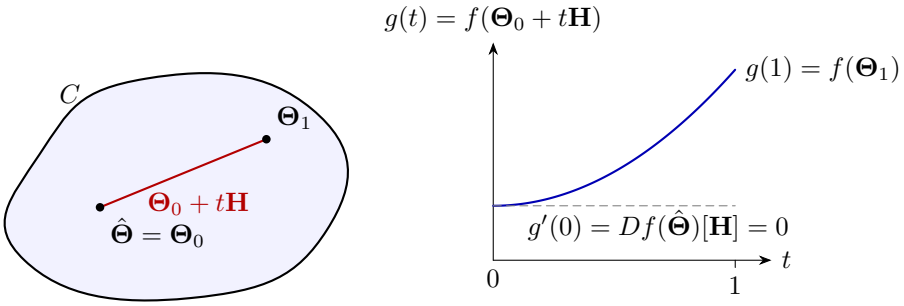


Figure. *The proof of Theorem 2.25: restricted to any segment from the critical point $\hat{\boldsymbol{\Theta}}$, the objective is a strictly convex function g with $g'(0) = 0$, so $g(1) > g(0)$. The open convex set C is, for example, the set of positive definite matrices.*

Theorem 2.26: transformations that keep the optimizer; profiling (Lab 3, Lemma 6.2, Proposition 6.3)

Let f be a function on a set Θ, $a > 0$ and c a constant not depending on θ.
(i) $\arg\min_\theta \{af(\theta) + c\} = \arg\min_\theta f(\theta)$.
(ii) $\arg\max_\theta f(\theta) = \arg\min_\theta \{-f(\theta)\}$.
(iii) $\arg\max_\theta L(\theta) = \arg\max_\theta \log L(\theta)$ for $L > 0$.
(iv) (Change of variables) If $\phi : \Theta \to \Xi$ is a bijection, $\arg\min_\theta f(\theta) = \phi^{-1}(\arg\min_\eta f(\phi^{-1}(\eta)))$.
(v) (Profiling) Let $\ell(\theta, \eta)$ be a function of two arguments. Suppose that for each η, $\hat\theta(\eta)$ minimizes $\ell(\cdot, \eta)$, and define the profile function $p(\eta) = \ell(\hat\theta(\eta), \eta)$. If $\hat\eta$ minimizes p, then $(\hat\theta(\hat\eta), \hat\eta)$ minimizes ℓ jointly.

Proof. For any $\theta, \theta_1 \in \Theta$:

(i) $af(\theta) + c \le af(\theta_1) + c$ iff $f(\theta) \le f(\theta_1)$, since $a > 0$;

(ii) $f(\theta) \ge f(\theta_1)$ iff $-f(\theta) \le -f(\theta_1)$;

(iii) $L(\theta) \ge L(\theta_1)$ iff $\log L(\theta) \ge \log L(\theta_1)$, since log is strictly increasing;

(iv) with $\eta = \phi(\theta)$, $\eta_1 = \phi(\theta_1)$, $f(\theta) \le f(\theta_1)$ iff $f(\phi^{-1}(\eta)) \le f(\phi^{-1}(\eta_1))$, and η_1 ranges over all of Ξ as θ_1 ranges over Θ;

(v) for any (θ, η), $\ell(\theta, \eta) \ge \ell(\hat\theta(\eta), \eta) = p(\eta) \ge p(\hat\eta) = \ell(\hat\theta(\hat\eta), \hat\eta)$. ■

The MLE (Ch. 4) first profiles out $\boldsymbol{\mu}$ and then minimizes over $\boldsymbol{\Sigma}$.

7 Multivariate Normal Distribution: the Facts Used Here

Definition 2.27: multivariate normal distribution (Lab 3, Definition 7.1, Proposition 7.2)

For $\boldsymbol{\mu} \in \mathbb{R}^p$ and $\boldsymbol{\Sigma} \succ 0$, $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ iff $\mathbf{X}$ has density $f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\}$, $\mathbf{x} \in \mathbb{R}^p$ (Part 1 slides, p. 123). For i.i.d. $Z_1, \dots, Z_p \sim N(0, 1)$, χ^2_p is the distribution of $Z_1^2 + \cdots + Z_p^2$. If $\mathbf{Z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, then (i) $\mathbf{t}^\top \mathbf{Z} \sim N(0, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$ for every $\mathbf{t} \in \mathbb{R}^p$ ($N(0, 0)$ is the point mass at 0), and (ii) $\mathbf{Z}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Z} \sim \chi^2_p$ (Results 4.2 and 4.7; proofs in Ch. 3).

8 Probability and Convergence

Motivation. Results 4.12 and 4.13 describe the limits of the sample mean and the sample covariance matrix; this section collects the tools they need: convergence of random vectors and matrices, the CLT, Slutsky's lemma and the Cramér–Wold device.

Definition 2.28: standard results from mathematical statistics (Lab 3, Recall 8.1)

Definitions and moments.

(S0) For random vectors, $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$ means $P(\|\mathbf{Y}_n - \mathbf{c}\| > \epsilon) \to 0$ for every $\epsilon > 0$, and $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ means $\mathrm{E}g(\mathbf{Y}_n) \to \mathrm{E}g(\mathbf{Y})$ for every bounded continuous g. $\mathrm{Cov}(\mathbf{Y}) = \mathrm{E}\{(\mathbf{Y} - \mathrm{E}\mathbf{Y})(\mathbf{Y} - \mathrm{E}\mathbf{Y})^\top\}$.

(S1) Markov: for a random variable $W \ge 0$ and $a > 0$, $P(W \ge a) \le \mathrm{E}(W)/a$.

(S2) $\mathrm{E}(\mathbf{AY}+\mathbf{b}) = \mathbf{A}\mathrm{E}\mathbf{Y}+\mathbf{b}$, $\mathrm{Cov}(\mathbf{AY}+\mathbf{b}) = \mathbf{A}\mathrm{Cov}(\mathbf{Y})\mathbf{A}^\top$; for independent $\mathbf{Y}_1,\dots,\mathbf{Y}_n$, $\mathrm{Cov}(\sum_j \mathbf{Y}_j) = \sum_j \mathrm{Cov}(\mathbf{Y}_j)$; $\mathrm{E}\|\mathbf{Y}-\mathrm{E}\mathbf{Y}\|^2 = \mathrm{tr}\mathrm{Cov}(\mathbf{Y})$; and $\mathrm{E}(V^2)=0$ implies $V=0$ almost surely.

Limit theorems.

(S3) Weak law of large numbers: if $Y_1, Y_2, \dots$ are i.i.d. with $\mathrm{E}|Y_1| < \infty$, $\frac{1}{n}\sum_{j=1}^n Y_j \xrightarrow{p} \mathrm{E}(Y_1)$.

(S4) Univariate CLT: if $Y_1, Y_2, \dots$ are i.i.d. with mean m and variance $v < \infty$, $\sqrt{n}(\bar{Y}_n - m) \xrightarrow{d} N(0, v)$.

Tools for limits.

(S5) Continuous mapping theorem (CMT): if $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ and g is continuous, $g(\mathbf{Y}_n) \xrightarrow{d} g(\mathbf{Y})$; if $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$ and g is continuous at $\mathbf{c}$, $g(\mathbf{Y}_n) \xrightarrow{p} g(\mathbf{c})$ (g need only be continuous at the single point $\mathbf{c}$).

(S6) Slutsky: if $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ and $\mathbf{W}_n \xrightarrow{p} \mathbf{c}$ for a constant $\mathbf{c}$, then $(\mathbf{Y}_n, \mathbf{W}_n) \xrightarrow{d} (\mathbf{Y}, \mathbf{c})$.

(S7) Joint convergence to constants: if $\mathbf{Y}_n^{(k)} \xrightarrow{p} \mathbf{c}_k$, $k = 1,\dots,m$, then $(\mathbf{Y}_n^{(1)},\dots,\mathbf{Y}_n^{(m)}) \xrightarrow{p} (\mathbf{c}_1,\dots,\mathbf{c}_m)$; a deterministic sequence $\mathbf{c}_n \to \mathbf{c}$ satisfies $\mathbf{c}_n \xrightarrow{p} \mathbf{c}$.

Theorem 2.29: Cauchy–Schwarz inequality for random variables (Lab 3, Lemma 8.2)

For random variables U, V with $\mathrm{E}(U^2) < \infty$ and $\mathrm{E}(V^2) < \infty$, $\mathrm{E}|UV| \le \{\mathrm{E}(U^2)\mathrm{E}(V^2)\}^{1/2}$.

Proof. Since $|UV| \le (U^2+V^2)/2$, $\mathrm{E}|UV| < \infty$. If $\mathrm{E}(V^2) = 0$, then $V = 0$ almost surely by (S2), and both sides are 0. Otherwise, for every $t \in \mathbb{R}$, $0 \le \mathrm{E}(|U| - t|V|)^2 = \mathrm{E}(U^2) - 2t\mathrm{E}|UV| + t^2\mathrm{E}(V^2)$. Taking $t = \mathrm{E}|UV|/\mathrm{E}(V^2)$ gives $0 \le \mathrm{E}(U^2) - \{\mathrm{E}|UV|\}^2/\mathrm{E}(V^2)$, which is the claim. ■

Theorem 2.30: vector Markov inequality and matrix convergence (Lab 3, Lemma 8.3, Definition 8.4, Lemma 8.5)

(i) For a random vector $\mathbf{Y}$ and $\epsilon > 0$, $P(\|\mathbf{Y}\| > \epsilon) \le \mathrm{E}\|\mathbf{Y}\|^2/\epsilon^2$.
(ii) For random $p \times q$ matrices $\mathbf{A}_n$ and a fixed $\mathbf{A}$, $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ means that every entry of $\mathbf{A}_n$ converges in probability to the corresponding entry of $\mathbf{A}$. Then $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ iff $\|\mathbf{A}_n - \mathbf{A}\|_F \xrightarrow{p} 0$ (Part 1 slides, p. 160).
(iii) Consequently $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ iff the vector of entries of $\mathbf{A}_n$ converges in probability, in the sense of (S0), to that of $\mathbf{A}$; hence (S5)–(S7) apply to random matrices.

Proof. (i) $\{\|\mathbf{Y}\| > \epsilon\} \subset \{\|\mathbf{Y}\|^2 \ge \epsilon^2\}$; apply (S1) to $W = \|\mathbf{Y}\|^2$ with $a = \epsilon^2$.

(ii) Suppose $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$. Since $\|\mathbf{A}_n - \mathbf{A}\|_F^2 \le pq \max_{ij}(\mathbf{A}_n - \mathbf{A})_{ij}^2$, the event $\|\mathbf{A}_n - \mathbf{A}\|_F > \epsilon$ implies $|(\mathbf{A}_n - \mathbf{A})_{ij}| > \epsilon/\sqrt{pq}$ for some (i,j), so $P(\|\mathbf{A}_n - \mathbf{A}\|_F > \epsilon) \le \sum_{ij} P(|(\mathbf{A}_n - \mathbf{A})_{ij}| > \epsilon/\sqrt{pq}) \to 0$. Conversely $|(\mathbf{A}_n - \mathbf{A})_{ij}| \le \|\mathbf{A}_n - \mathbf{A}\|_F$ for every (i,j).

(iii) $\|\mathbf{A}_n - \mathbf{A}\|_F$ equals the Euclidean norm of the difference of the vectors of entries, so (iii) is (ii) restated with (S0). ■

Theorem 2.31: characteristic functions and the Cramér–Wold device (Lab 3, Theorem 8.6, Corollary 8.7)

For a random vector $\mathbf{Y} \in \mathbb{R}^k$ let $\varphi_{\mathbf{Y}}(\mathbf{t}) = \mathrm{E}\exp(i\mathbf{t}^\top \mathbf{Y})$, $\mathbf{t} \in \mathbb{R}^k$.
(i) (Uniqueness) $\varphi_{\mathbf{Y}}(\mathbf{t}) = \varphi_{\mathbf{W}}(\mathbf{t})$ for all $\mathbf{t}$ iff $\mathbf{Y}$ and $\mathbf{W}$ have the same distribution.
(ii) (Lévy's continuity theorem) $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ iff $\varphi_{\mathbf{Y}_n}(\mathbf{t}) \to \varphi_{\mathbf{Y}}(\mathbf{t})$ for all $\mathbf{t} \in \mathbb{R}^k$.
These are proved in mathematical statistics (Part 1 slides, p. 125). **Cramér–Wold device**:
(iii) the distribution of $\mathbf{Y}$ is determined by the distributions of $\mathbf{t}^\top \mathbf{Y}$, $\mathbf{t} \in \mathbb{R}^k$: if $\mathbf{t}^\top \mathbf{Y}$ and $\mathbf{t}^\top \mathbf{W}$ have the same distribution for every $\mathbf{t}$, so do $\mathbf{Y}$ and $\mathbf{W}$;
(iv) $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ iff $\mathbf{t}^\top \mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top \mathbf{Y}$ for every $\mathbf{t} \in \mathbb{R}^k$.

Proof of (iii) and (iv). For every $\mathbf{t}$, the definition gives the key identity $\varphi_{\mathbf{t}^\top \mathbf{Y}}(1) = \mathrm{E}e^{i\mathbf{t}^\top \mathbf{Y}} = \varphi_{\mathbf{Y}}(\mathbf{t})$.

(iii) If $\mathbf{t}^\top \mathbf{Y} \overset{d}{=} \mathbf{t}^\top \mathbf{W}$, then $\varphi_{\mathbf{Y}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top \mathbf{Y}}(1) = \varphi_{\mathbf{t}^\top \mathbf{W}}(1) = \varphi_{\mathbf{W}}(\mathbf{t})$. This holds for all $\mathbf{t}$, so (i) gives the claim.

(iv) If $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, then $\mathbf{t}^\top \mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top \mathbf{Y}$ by the CMT (S5). Conversely, if $\mathbf{t}^\top \mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top \mathbf{Y}$, (ii) applied to these scalars gives $\varphi_{\mathbf{t}^\top \mathbf{Y}_n}(s) \to \varphi_{\mathbf{t}^\top \mathbf{Y}}(s)$ for every $s \in \mathbb{R}$; at $s = 1$, $\varphi_{\mathbf{Y}_n}(\mathbf{t}) \to \varphi_{\mathbf{Y}}(\mathbf{t})$. Since $\mathbf{t}$ is arbitrary, (ii) gives $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$. ■

"All one-dimensional linear projections determine the multivariate distribution" underlies the general definition of the multivariate normal (Ch. 3) and the multivariate CLT.

9 Selected exercises

Motivation. Three further tools: the min–max characterization of every eigenvalue (which makes sample eigenvalues consistent), Hadamard's determinant inequality (which bounds $|\mathbf{R}|$ in independence tests), and the Kronecker product (the language for the covariance of a whole data matrix).

Exercise 2.1 (Courant–Fischer and Weyl)

[Supplement] Let $\mathbf{A}$ be symmetric $p \times p$ with eigenvalues $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$ and orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_p$.
(a) Show that for $k = 1, \ldots, p$,

$$\lambda_k(\mathbf{A}) = \max_{\mathcal{V}:\ \dim \mathcal{V} = k} \ \min_{\mathbf{x} \in \mathcal{V},\ \|\mathbf{x}\| = 1} \mathbf{x}^\top \mathbf{A} \mathbf{x}.$$

(b) For symmetric $\mathbf{A}, \mathbf{B}$ show $|\lambda_k(\mathbf{A}) - \lambda_k(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|_F$ for every k, which extends Theorem 2.10(ii) to all eigenvalues.

Solution [Supplement].

(a)

Step 1 ($\geq$). Let $\mathcal{V}_k = \mathrm{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$. A unit vector $\mathbf{x} \in \mathcal{V}_k$ is $\mathbf{x} = \sum_{i \leq k} c_i \mathbf{u}_i$ with $\sum_i c_i^2 = 1$, so $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i \leq k} \lambda_i c_i^2 \geq \lambda_k$. Hence the minimum over $\mathcal{V}_k$ is at least λ_k (it equals λ_k at $\mathbf{x} = \mathbf{u}_k$), and the maximum over all $\mathcal{V}$ is at least λ_k.

Step 2 ($\leq$). Let $\mathcal{V}$ be any subspace with $\dim \mathcal{V} = k$ and $\mathcal{W} = \mathrm{span}(\mathbf{u}_k, \ldots, \mathbf{u}_p)$, $\dim \mathcal{W} = p - k + 1$. Since $\dim \mathcal{V} + \dim \mathcal{W} = p + 1 > p$, the intersection $\mathcal{V} \cap \mathcal{W}$ contains a unit vector $\mathbf{x} = \sum_{i \geq k} c_i \mathbf{u}_i$, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i \geq k} \lambda_i c_i^2 \leq \lambda_k$. So the minimum over $\mathcal{V}$ is at most λ_k for every $\mathcal{V}$.

(b)

Step 1. Let $\epsilon = \|\mathbf{A}-\mathbf{B}\|_F$. As in the proof of Theorem 2.10(ii), $|\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{B}\mathbf{x}| \le \epsilon$ for every unit $\mathbf{x}$.

Step 2. Fix $\mathcal{V}$ with $\dim\mathcal{V} = k$. The unit sphere of $\mathcal{V}$ is compact, so $\mathbf{x}\mapsto\mathbf{x}^\top\mathbf{B}\mathbf{x}$ attains its minimum on it at some $\mathbf{x}_0$. Then

$$\min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1}\mathbf{x}^\top\mathbf{A}\mathbf{x} \le \mathbf{x}_0^\top\mathbf{A}\mathbf{x}_0 \le \mathbf{x}_0^\top\mathbf{B}\mathbf{x}_0 + \epsilon = \min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1}\mathbf{x}^\top\mathbf{B}\mathbf{x} + \epsilon.$$

Step 3. Taking the maximum over $\mathcal{V}$ and using (a) gives $\lambda_k(\mathbf{A}) \le \lambda_k(\mathbf{B}) + \epsilon$. Exchanging $\mathbf{A}$ and $\mathbf{B}$ gives $\lambda_k(\mathbf{B}) \le \lambda_k(\mathbf{A}) + \epsilon$.

Consequently each $\lambda_k(\cdot)$ is continuous, and $\mathbf{S}\xrightarrow{p}\boldsymbol{\Sigma}$ implies $\lambda_k(\mathbf{S})\xrightarrow{p}\lambda_k(\boldsymbol{\Sigma})$ for every k (sample PC variances are consistent). ■

Exercise 2.2 (Hadamard's inequality)

[Supplement] Let $\mathbf{A}\succ 0$ be $p\times p$. Show that $|\mathbf{A}| \le \prod_{i=1}^p a_{ii}$, with equality iff $\mathbf{A}$ is diagonal. Deduce that a correlation matrix $\mathbf{R}\succ 0$ satisfies $0 < |\mathbf{R}| \le 1$, with $|\mathbf{R}| = 1$ iff $\mathbf{R} = \mathbf{I}_p$.

Solution [Supplement].

Step 1 (reduction). $a_{ii} = \mathbf{e}_i^\top\mathbf{A}\mathbf{e}_i > 0$. Let $\mathbf{D} = \mathrm{diag}(a_{11},\dots,a_{pp})$ and $\mathbf{B} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. Then $\mathbf{B}\succ 0$, $b_{ii} = 1$ and $|\mathbf{B}| = |\mathbf{A}|/\prod_i a_{ii}$, so it suffices to show $|\mathbf{B}| \le 1$ with equality iff $\mathbf{B} = \mathbf{I}_p$.

Step 2. Let $\mu_1,\dots,\mu_p > 0$ be the eigenvalues of $\mathbf{B}$. By Theorem 2.9, $\sum_i\mu_i = \mathrm{tr}\mathbf{B} = p$ and $\log|\mathbf{B}| = \sum_i\log\mu_i$. Theorem 2.15 gives $\log\mu_i \le \mu_i - 1$, so

$$\log|\mathbf{B}| \le \sum_{i=1}^p(\mu_i - 1) = \mathrm{tr}\mathbf{B} - p = 0.$$

Step 3 (equality). Equality holds iff $\mu_i = 1$ for all i, i.e. $\mathbf{B} = \mathbf{U}\mathbf{I}_p\mathbf{U}^\top = \mathbf{I}_p$, i.e. $\mathbf{A} = \mathbf{D}$ is diagonal.

For a correlation matrix $a_{ii} = 1$, and $|\mathbf{R}| > 0$ because $\mathbf{R}\succ 0$. ■

Exercise 2.3 (Kronecker products and vec)

[Supplement] For $\mathbf{A}\in\mathbb{R}^{m\times n}$ and $\mathbf{B}\in\mathbb{R}^{r\times s}$ let $\mathbf{A}\otimes\mathbf{B} = [a_{ij}\mathbf{B}]_{i,j}\in\mathbb{R}^{mr\times ns}$, and let $\mathrm{vec}(\mathbf{X})$ stack the columns of $\mathbf{X}$.
(a) Show $(\mathbf{A}\otimes\mathbf{B})(\mathbf{C}\otimes\mathbf{D}) = \mathbf{A}\mathbf{C}\otimes\mathbf{B}\mathbf{D}$ whenever the products are defined, $(\mathbf{A}\otimes\mathbf{B})^\top = \mathbf{A}^\top\otimes\mathbf{B}^\top$, and $(\mathbf{A}\otimes\mathbf{B})^{-1} = \mathbf{A}^{-1}\otimes\mathbf{B}^{-1}$ for invertible $\mathbf{A},\mathbf{B}$.
(b) For symmetric $\mathbf{A}$ $(m\times m)$ and $\mathbf{B}$ $(r\times r)$ with eigenpairs $(\lambda_i,\mathbf{u}_i)$ and $(\mu_j,\mathbf{v}_j)$, show that $\mathbf{A}\otimes\mathbf{B}$ has the orthonormal eigenvectors $\mathbf{u}_i\otimes\mathbf{v}_j$ with eigenvalues $\lambda_i\mu_j$; hence $|\mathbf{A}\otimes\mathbf{B}| = |\mathbf{A}|^r|\mathbf{B}|^m$ and $\mathrm{tr}(\mathbf{A}\otimes\mathbf{B}) = \mathrm{tr}\mathbf{A}\,\mathrm{tr}\mathbf{B}$.
(c) Show $\mathrm{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\mathbf{B}^\top\otimes\mathbf{A})\,\mathrm{vec}(\mathbf{X})$.
(d) If the $n\times p$ data matrix $\mathbf{X}$ has iid rows $N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$, show $\mathrm{vec}(\mathbf{X})\sim N_{np}(\boldsymbol{\mu}\otimes\mathbf{1}_n,\ \boldsymbol{\Sigma}\otimes\mathbf{I}_n)$.

Solution [Supplement].

(a) The (i,k) block of the product is $\sum_j(a_{ij}\mathbf{B})(c_{jk}\mathbf{D}) = (\sum_j a_{ij}c_{jk})\mathbf{B}\mathbf{D} = (\mathbf{A}\mathbf{C})_{ik}\mathbf{B}\mathbf{D}$, the (i,k) block of $\mathbf{A}\mathbf{C}\otimes\mathbf{B}\mathbf{D}$. The (i,j) block of $(\mathbf{A}\otimes\mathbf{B})^\top$ is the transpose of the (j,i) block, $a_{ji}\mathbf{B}^\top$. Finally $(\mathbf{A}\otimes\mathbf{B})(\mathbf{A}^{-1}\otimes\mathbf{B}^{-1}) = \mathbf{I}_m\otimes\mathbf{I}_r = \mathbf{I}_{mr}$.

(b)

Step 1. Treating vectors as one-column matrices, (a) gives $(\mathbf{A}\otimes\mathbf{B})(\mathbf{u}_i\otimes\mathbf{v}_j) = \mathbf{A}\mathbf{u}_i\otimes\mathbf{B}\mathbf{v}_j = \lambda_i\mu_j\,\mathbf{u}_i\otimes\mathbf{v}_j$.

Step 2. $(\mathbf{u}_i \otimes \mathbf{v}_j)^\top (\mathbf{u}_k \otimes \mathbf{v}_l) = (\mathbf{u}_i^\top \mathbf{u}_k)(\mathbf{v}_j^\top \mathbf{v}_l) = \delta_{ik}\delta_{jl}$, so these mr vectors form an orthonormal eigenbasis of the symmetric matrix $\mathbf{A} \otimes \mathbf{B}$.

Step 3. By Theorem 2.9, $|\mathbf{A} \otimes \mathbf{B}| = \prod_{i,j} \lambda_i \mu_j = (\prod_i \lambda_i)^r (\prod_j \mu_j)^m = |\mathbf{A}|^r |\mathbf{B}|^m$ and $\operatorname{tr}(\mathbf{A} \otimes \mathbf{B}) = \sum_{i,j} \lambda_i \mu_j = \operatorname{tr}\mathbf{A} \operatorname{tr}\mathbf{B}$.

(c) Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have columns $\mathbf{x}_1, \dots, \mathbf{x}_p$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$. Column k of $\mathbf{AXB}$ is $\mathbf{AXb}_k = \sum_j b_{jk} \mathbf{A}\mathbf{x}_j = [b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}] \operatorname{vec}(\mathbf{X})$, and $[b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}]$ is the kth block row of $\mathbf{B}^\top \otimes \mathbf{A}$.

(d) Column j of $\mathbf{X}$ is $\mathbf{x}_{(j)} = (X_{1j}, \dots, X_{nj})^\top$ with mean $\mu_j \mathbf{1}_n$, so $\mathrm{E} \operatorname{vec}(\mathbf{X}) = \boldsymbol{\mu} \otimes \mathbf{1}_n$. For columns j, k, $\operatorname{Cov}(X_{ij}, X_{lk}) = \sigma_{jk}\delta_{il}$ by independence of the rows, so $\operatorname{Cov}(\mathbf{x}_{(j)}, \mathbf{x}_{(k)}) = \sigma_{jk}\mathbf{I}_n$, the (j,k) block of $\boldsymbol{\Sigma} \otimes \mathbf{I}_n$. Normality: the stacked rows $(\mathbf{X}_1^\top, \dots, \mathbf{X}_n^\top)^\top$ are jointly normal (independent normal blocks), and $\operatorname{vec}(\mathbf{X})$ is a permutation, hence a linear map, of this vector (Theorem 3.8). ■